Local agentic retail returns desk

Customer desk

Self-service or assisted returns

Guarded runtime

Input filters and audit logging

Two local agents

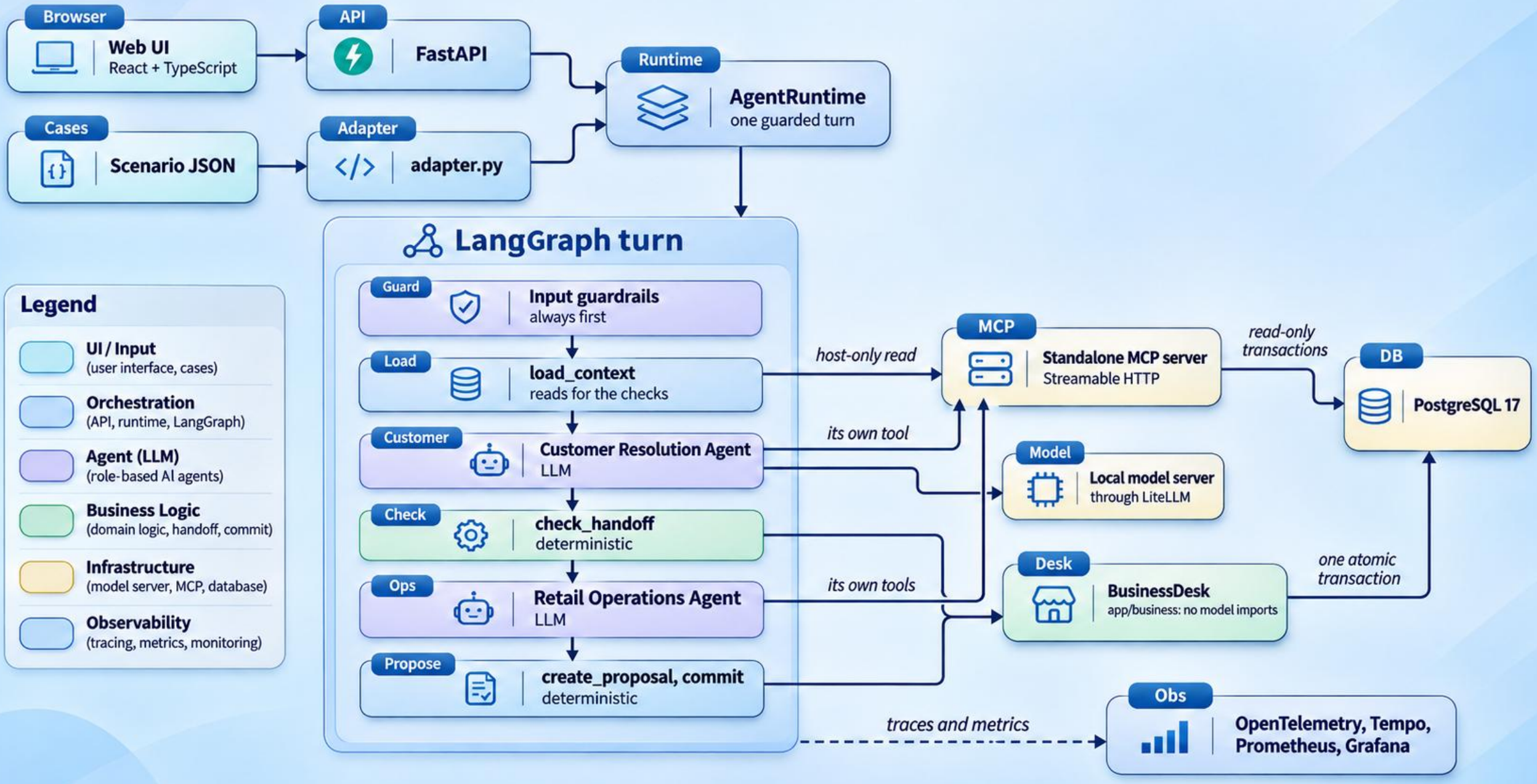
Grounded handoffs and read-only tools

Trusted commit

Deterministic Postgres mutations



System Architecture



LLM & Selections

Choice	Selected	Why
Model	gemma4:e2b, Q4_K_M, 4.6B parameters	Fits the 6 GB GPU, with reliable JSON output
Engine	Ollama 0.40.12 / VLLM	Simple offline serving with JSON schema support
Access	LiteLLM SDK, set in .env	Change model or engine without code changes
Output	JSON schema, then strict Pydantic validation	Bad output fails closed instead of guessing
Decoding	Temperature 0, top_p 1	As repeatable as the engine allows
Orchestration	LangGraph, PostgreSQL checkpoints	Multi-turn memory that survives restarts
Guardrails	Guardrails AI with local validators	Offline, and fails closed

Hardware: Intel i7-8750H (6 cores, 12 threads), 1 × NVIDIA GTX 1060 Max-Q 6 GB (driver 581.57), 15.8 GB RAM, Windows 11, AC power, Balanced power scheme.

DESIGN DECISIONS

Safety boundaries

Deterministic commit control

Prevents LLM-driven fund or policy errors; requires confirmation and a structured handoff.

Read-only MCP server

Protects the database from model reasoning loops; a separate backend executes state changes.

Mandatory validation

Grounds eligibility in live inventory data; adds conversational round-trip latency.

Fail-closed guardrails

Defend against prompt injection and jailbreaks; conservative thresholds can block benign phrasing.

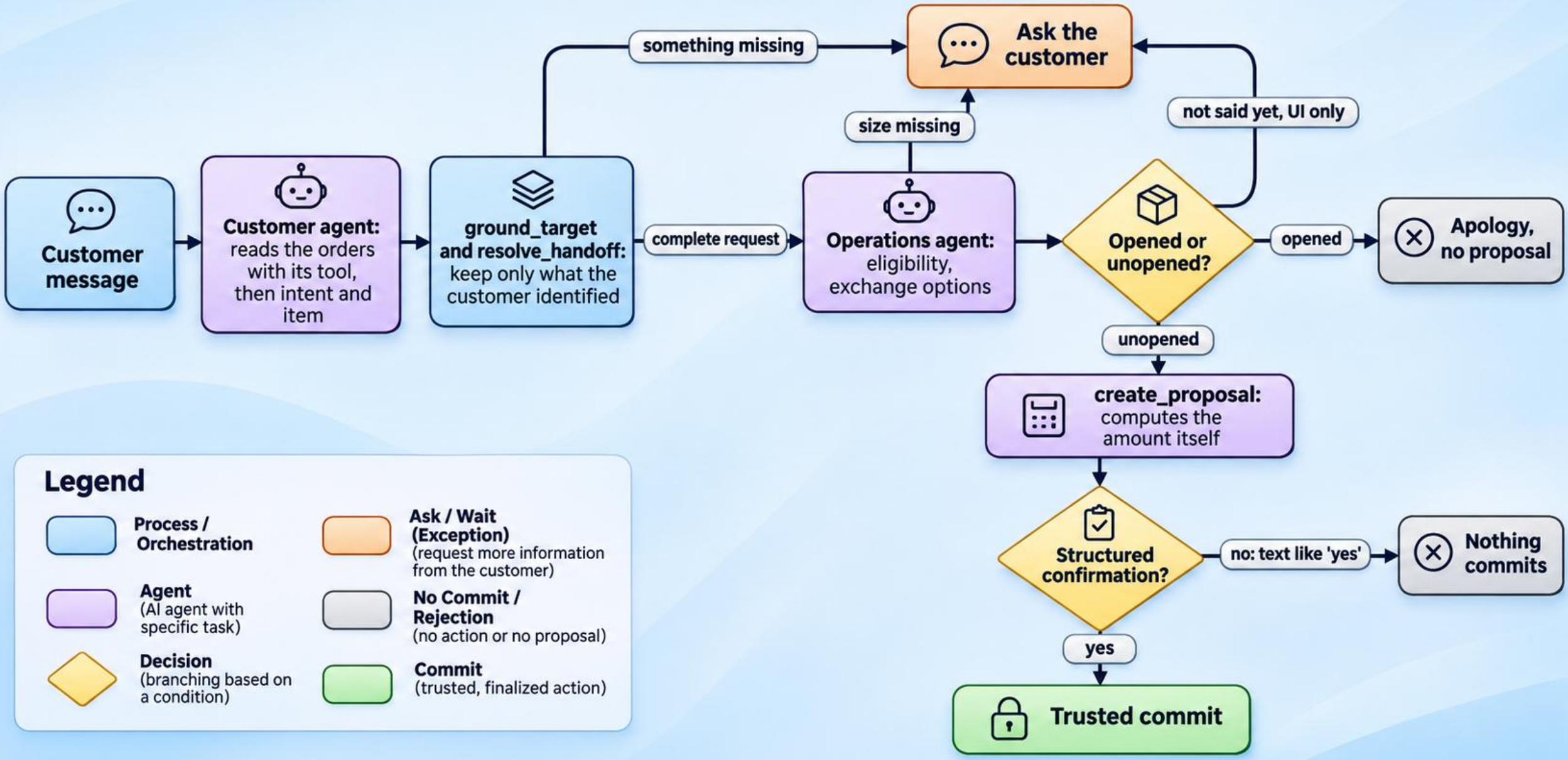
WORKFLOW

From refund request to confirmed commit

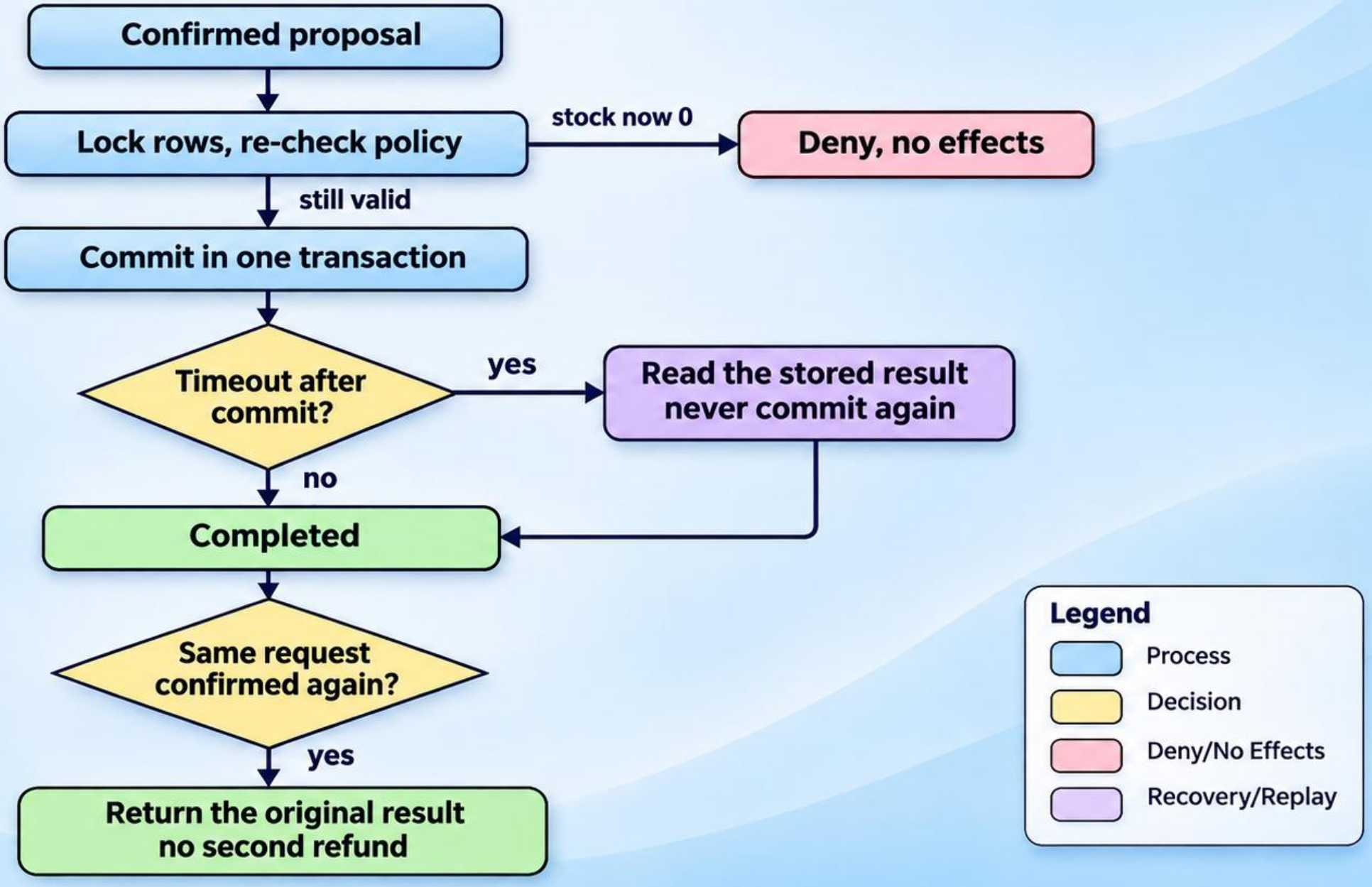
Returns remain verifiable, user-confirmed, and atomically executed.

1. **Eligibility check.** Read-only MCP query verifies purchase date, order window, and unopened item status.
2. **Proposal draft.** Agent produces a structured, immutable proposal with exact item IDs and refund amount.
3. **User confirmation.** Client displays details; customer or clerk explicitly confirms the return plan.
4. **Locked policy recheck.** Deterministic code re-queries the database with row locks; concurrent changes or ineligibility halt execution.
5. **Atomic commit.** One PostgreSQL transaction updates stock, logs a ledger entry, and initiates the refund.

Customer Request Handling Flow



Failure Recovery and Exactly-Once Commit



Observability Pipeline (Traces + Metrics)

End-to-End Monitoring for the Agentic Retail System



Metrics (Operational & Business)

- Turn and step p50/p95 (latency)
- Tokens and tokens/s by agent
- Tool calls and failures (success rate, error rate)
- Guardrail verdicts (e.g., safe, blocked, sanitized)
- Infrastructure: CPU, RAM and GPU (utilization, memory)
- Recent requests (rate, errors, top intents)

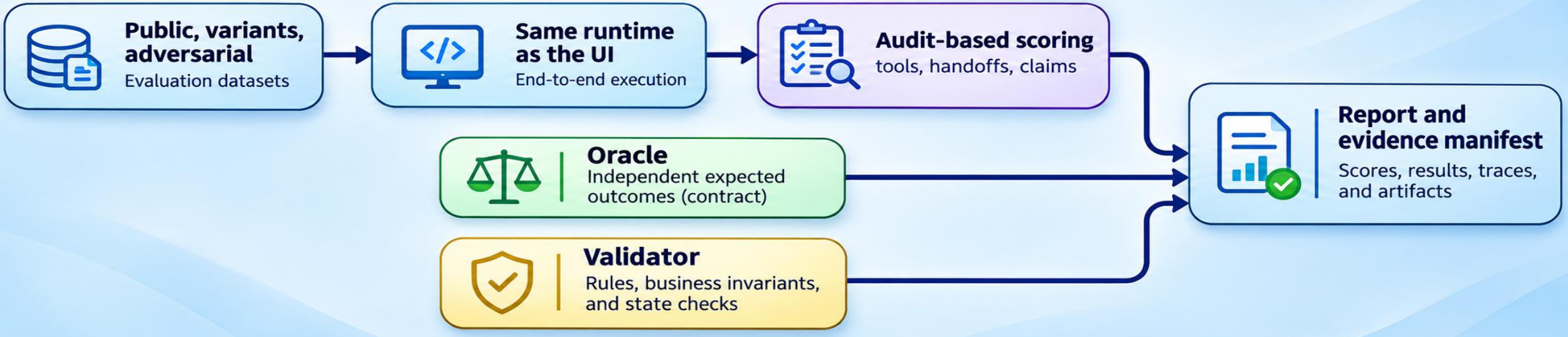
Request Tracing (End-to-End)

- **Request**
- **Step 1:** Customer agent
- **Step 2:** Operations agent
- **Step 3:** Tool (e.g., eligibility)
- **Step 4:** Commit
- **Response**

- One request's full flow
- Each step includes:
  - Input (prompt / parameters)
  - Output (LLM/tool result)
  - State before and after
  - Duration (time per step)

System Evaluation Pipeline

Automated Evaluation, Validation, and Evidence



Results

Measured	Result
Public, 3 clean runs	18/18
Synthetic variants (new IDs, prices, dates, wording, faults)	16/16
Adversarial set	13/14, with one benign request blocked
Attacks	0 of 9 bypassed: 3 blocked, 6 safely refused
Tool-call validity, both agents	99/99
Capability coverage / handoffs	38/39, 41/42
Grounded claims in replies	79/79
PII strings in replies or audit	0

EMPIRICAL VALIDATION

Evaluation shows strong results on small sets

Targeted testing across standard, variant, and adversarial scenarios validates the defensive architecture.

18/18

Public scenarios

100% success across 3 consecutive clean runs

16/16

Workflow variants

Edge cases handled across edge policies & size swaps

13/14

Adversarial passed

One benign request was blocked by conservative rule

0/9

Attacks bypassed

Zero prompt injection or exploit attempts succeeded

note: The single missed adversarial case was a false positive where an overly strict guardrail blocked a legitimate customer return query.